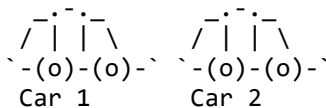


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# Equation of Motion: Lecture 5

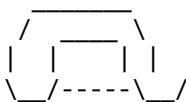
## Advanced Problem Solving & Galileo's Trick

### Problem 1: Car Race Analysis



- **Description:** Two cars start from rest and race over the same distance.
- **Given Information:**
  - Both cars start from *rest* (initial velocity  $u = 0$ ).
  - Car 1 has constant acceleration  $a_1$ .
  - Car 2 has constant acceleration  $a_2$ .
  - Car 1 completes the race  $t$  seconds earlier than Car 2 (meaning  $t_2 - t_1 = t$ ).
  - The difference in their final velocities is  $V$  ( $V_1 - V_2 = V$ ).
- **Goal:** Prove the relation  $V/t = \sqrt{a_1 a_2}$ .
- **Solution Steps:**
  1. Use the equation  $s = ut + (1/2)at^2$  for both cars (since  $u=0$ ,  $s = (1/2)at^2$ ).
    - For Car 1:  $s = (1/2)a_1 t_1^2 \Rightarrow t_1 = \sqrt{(2s/a_1)}$ .
    - For Car 2:  $s = (1/2)a_2 t_2^2 \Rightarrow t_2 = \sqrt{(2s/a_2)}$ .
  2. Use the equation  $v^2 = u^2 + 2as$  for both cars (since  $u=0$ ,  $v = \sqrt{(2as)}$ ).
    - For Car 1:  $v_1 = \sqrt{(2a_1 s)}$ .
    - For Car 2:  $v_2 = \sqrt{(2a_2 s)}$ .
  3. Substitute the expressions for  $t_1, t_2$  into  $t_2 - t_1 = t$ :
    - $t = \sqrt{(2s/a_2)} - \sqrt{(2s/a_1)}$ .
    - $t = \sqrt{2s} * (1/\sqrt{a_2} - 1/\sqrt{a_1})$ .
    - $t = \sqrt{2s} * (\sqrt{a_1} - \sqrt{a_2}) / \sqrt{(a_1 a_2)}$ .
  4. Substitute the expressions for  $v_1, v_2$  into  $V_1 - V_2 = V$ :
    - $V = \sqrt{(2a_1 s)} - \sqrt{(2a_2 s)}$ .
    - $V = \sqrt{2s} * (\sqrt{a_1} - \sqrt{a_2})$ .
  5. Divide the equation for  $V$  by the equation for  $t$ :
    - $V/t = [\sqrt{2s} * (\sqrt{a_1} - \sqrt{a_2})] / [\sqrt{2s} * (\sqrt{a_1} - \sqrt{a_2}) / \sqrt{(a_1 a_2)}]$ .
    - $V/t = \sqrt{(a_1 a_2)}$ .

### Problem 2: Car with Constant Acceleration



- **Description:** A car moves with constant acceleration.
- **Given Information:**

- Covers a distance of **60 meters** in **6 seconds**.
- Its final velocity after these 6 seconds is **15 m/s**.
- **Goal:**
  1. Find the initial velocity ( **$u$** ).
  2. Find the constant acceleration ( **$a$** ).
  3. Find the distance the car would have been **behind** its starting point if it had started from rest with the same acceleration, reaching the initial velocity  **$u$** .
- **Solution Steps:**
  1. **Finding  $u$  and  $a$ :**
    - Using  **$v = u + at$** :  **$15 = u + a(6)$**  (Equation 1).
    - Using  **$s = ut + (1/2)at^2$** :  **$60 = u(6) + (1/2)a(6)^2$**   $\Rightarrow$   **$60 = 6u + 18a$**  (Equation 2).
    - Solve Equation 1 and Equation 2 simultaneously:
      - From (1),  **$u = 15 - 6a$** .
      - Substitute into (2):  **$60 = 6(15 - 6a) + 18a$**   $\Rightarrow$   **$60 = 90 - 36a + 18a$**   $\Rightarrow$   **$60 = 90 - 18a$** .
      - **$18a = 30$**   $\Rightarrow$   **$a = 30/18 = 5/3 \text{ m/s}^2$** .
      - Substitute  **$a$**  back into (1):  **$15 = u + 6(5/3)$**   $\Rightarrow$   **$15 = u + 10$**   $\Rightarrow$   **$u = 5 \text{ m/s}$** .
    - **Result:** **Initial velocity ( $u$ ) = 5 m/s**, **Acceleration ( $a$ ) = 5/3 m/s<sup>2</sup>**.
  2. **Finding distance if starting from rest:**
    - Initial velocity for this part:  **$u_{\text{rest}} = 0$** .
    - Final velocity:  **$v_{\text{final}} = 5 \text{ m/s}$**  (the initial velocity found in Part 1).
    - Acceleration:  **$a = 5/3 \text{ m/s}^2$** .
    - Using  **$v^2 = u^2 + 2as$** :  **$5^2 = 0^2 + 2(5/3)s$** .
    - **$25 = (10/3)s$**   $\Rightarrow$   **$s = 25 \cdot (3/10) = 75/10 = 7.5 \text{ meters}$** .
  3. **Result:** The car would have been **7.5 meters** behind the starting point if it had started from rest.

## Galileo's Trick (Law of Odd Numbers)



- **Concept:** This trick applies to the motion of objects starting from **rest** and moving with **constant acceleration**.
- **Principle:** In successive equal intervals of time, the **distances covered** by such an object are in the ratio of odd numbers:  **$1x : 3x : 5x : 7x \dots$** .
- **Conditions for Application:**
  - The object must start from **rest** (**initial velocity  $u = 0$** ).
  - The object must move with **constant acceleration**.
- **Example (Free Fall under Gravity):**
  - If a ball is dropped (starts from rest, accelerates by  **$g$** ).
  - Distance covered in the **1st second** is  **$x$** .
  - Distance covered in the **2nd second** is  **$3x$** .
  - Distance covered in the **3rd second** is  **$5x$** .
  - And so on...
- **Mathematical Relation:**
  - The total distance covered in  **$n$**  equal time intervals is proportional to  **$n^2$** .

- The sum of the first  $n$  odd numbers is  $n^2$  (e.g.,  $1+3=4=2^2$ ,  $1+3+5=9=3^2$ ), which aligns with this principle.
- **Graphical Interpretation:**
  - The area under a **Velocity-Time (V-T)** graph (which is a straight line starting from the origin for constant acceleration from rest) also follows this pattern.

### Problem 3: Dropping Ball - Galileo's Trick Application



- **Description:** A ball is dropped with zero initial velocity ( $u=0$ ) under gravity (constant acceleration).
- **Goal:** Find the ratio of displacement in the 1st, 2nd, and 3rd seconds.
- **Solution:**
  - Directly applying Galileo's Trick:
  - Displacement in the 1st second =  $x$ .
  - Displacement in the 2nd second =  $3x$ .
  - Displacement in the 3rd second =  $5x$ .
  - **Ratio:**  $1 : 3 : 5$ .
- **Advanced Application Example:**
  - To find the displacement in the  $n$ th second, the formula is  $(2n - 1)x$ .
  - **Displacement in the 100th second:**  $(2 \cdot 100 - 1)x = 199x$ .
  - **Ratio of displacement in 100th second to 1st second:**  $199x / x = 199$ .

### Problem 4: Dropping Water Drops - Galileo's Trick Application



- **Description:** Water drops fall at regular intervals from a tap.
- **Given Information:**
  - When the 1st drop hits the ground, the 3rd drop is just leaving the tap.
  - The total height from which the drops fall is 5 meters.
- **Goal:** Find the height of the 2nd drop above the ground.
- **Solution Strategy (using Galileo's Trick):**
  1. Let  $T$  be the regular time interval between drops.
  2. When the 1st drop hits the ground, it has been falling for  $2T$  seconds (since the 3rd drop is just leaving, meaning 2 intervals have passed).
  3. The 2nd drop has been falling for  $T$  seconds.
  4. Apply Galileo's Trick:
    - Distance covered by the first drop in the first time interval  $T = x$ .
    - Distance covered by the first drop in the second time interval  $T = 3x$ .
    - Therefore, the **total height** from which the drops fall is  $x + 3x = 4x$ .

5. Given that the *total height is 5 meters*:  $4x = 5 \text{ meters} \Rightarrow x = 5/4 \text{ meters}$ .
  6. The *second* drop has traveled for  $T$  seconds, covering a distance of  $x$ .
  7. The *height of the second drop above the ground* is *Total Height - Distance traveled by second drop*.
  8. Height above ground =  $4x - x = 3x$ .
  9. Calculate the value:  $3x = 3 \cdot (5/4) = 15/4 = 3.75 \text{ meters}$ .
- **Alternative Method (using Equations of Motion):**
    1. Let the total height be  $H = 5\text{m}$ . Let  $T$  be the time interval between drops.
    2. The first drop falls for  $2T$  seconds to cover  $H$ . So,  $H = (1/2)g(2T)^2 = 2gT^2$ .
    3. Since  $H = 5\text{m}$ ,  $5 = 2gT^2 \Rightarrow gT^2 = 2.5$ .
    4. The second drop falls for  $T$  seconds. The distance it covers from the tap is  $s_2 = (1/2)gT^2$ .
    5. Substitute  $gT^2 = 2.5$ :  $s_2 = (1/2)(2.5) = 1.25 \text{ meters}$ .
    6. The height of the second drop above the ground is  $H - s_2$ .
    7. Height above ground =  $5 - 1.25 = 3.75 \text{ meters}$ .
  - **Result**: The second drop is  $3.75 \text{ meters}$  above the ground.

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